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REVISION SERIES Coordination Compounds

NOTE

A dedicated Revision playlist link is in the description box of this chapter.
Containing :
-Revision Video
-2022 PYQs
-2023 PYQs

To get Notes + PYQs of this chapter

Link in description or go to www.dexterchem.in/courses to find PYQ and Revision Notes.

Coordination Compounds

Types of salts

Simple salt

NaCl, KCl

$\downarrow H_2O$

$Na^{\oplus} Cl^{\ominus}$, $K^{\oplus} Cl^{\ominus}$

Double salt

$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ → Potash Alum

$\downarrow H_2O$

$K^{\oplus}$, SO_4^{2-}, Al^{+3}

Co-ordination complexes.

$K_4[Fe(CN)_6]$

$\downarrow H_2O$

$4K^{\oplus} + [\overset{+2}{Fe}(CN)_6]^{4-}$

$K_4 [Fe(CN)_6]$

- $[Fe(CN)_6]$ → co-ordination sphere
- 6 → co-ordination
- CN → ligands / secondary valency.
- K_4 → Primary valency.
- Fe → central metal atom/ion

Central metal ion :-

The metal ion which form complex with anion, neutral molecule or positively charged ion is known as central metal ion (CMI).

* In general central metal ion acts as electron pair acceptor & forms co-ordinate bond with ligands.

Some times it also donates the e^-s to the ligand. (As in Back Bonding) $\boxed{\text{(synergic Bonding)}}$

Co-ordination sphere :-

The central metal ion (CMI) & the ligands directly attached to it are collectively termed as co-ordination sphere.

It is written in square brackets.

* The part outside the bracket is known as primary valency or ionizable valency. $K_4[Fe(CN)_6]$. $[Cu(NH_3)_4]SO_4$

Primary valency (for K_4)

Primary valency (for SO_4)

Ligand :-

Neutral / positively charged / negatively charged ion or molecules which combines with CMI to form complex are called ligands.

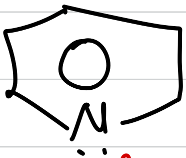
Classification of ligands

- Denticity (Ligancy)
- Charge
- electron donation or accepting tendency

On the basis of dentecy or ligancy :-

① **Monodentate ligand :-** which can donate 1e⁻ pair at a time.

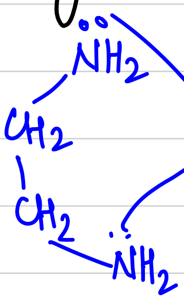
ex: $\ddot{Cl}^-$, Br^-, I^-, $\bar{C}N$, $\bar{O}H$, $\bar{S}H$, $\ddot{N}H_3$, $H_2\ddot{O}$, CO,

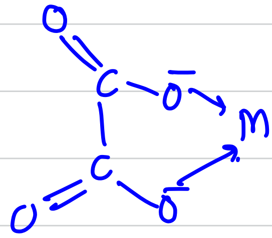

, $H^\ominus$ etc.

Pyridine hydride.

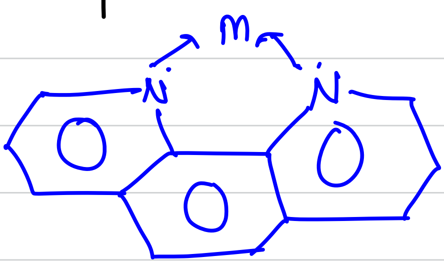
② **Bidentate ligand :-** which can donate 2e⁻ pair from two different atom/sites at a time

Imp eg. (a) ethylenediamine (en) ⓑ oxalate (ox)

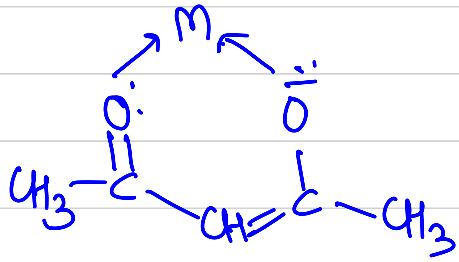




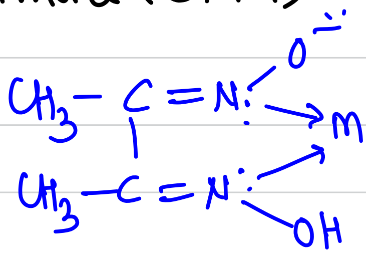
(c) 1,10-phenanthroline (o-phen)



(d) acetylacetonate (acac)



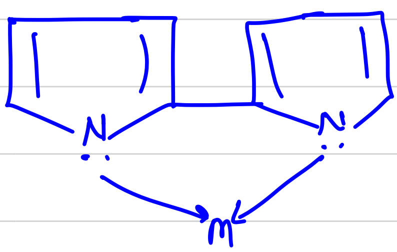
Imp (e) dimethylglyoximate (DMG)



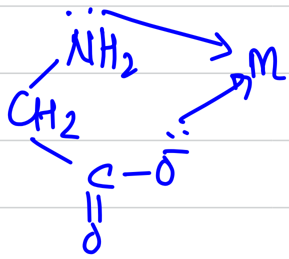
Imp

(donor atom is nitrogen & not oxygen)

(f) 2,2 dipyridyl (dipy)



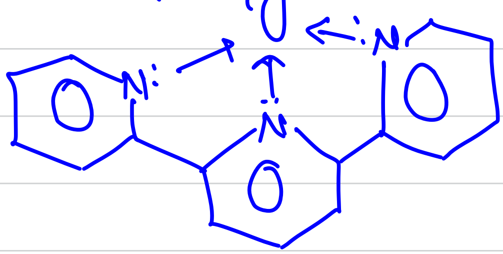
(g) glycinate (gly)



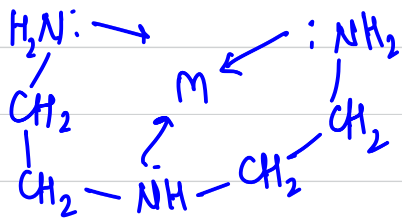
③ Tridentate ligand :-

3 lone pairs at a time from different atoms.

(a) Terpyridine (terpy)



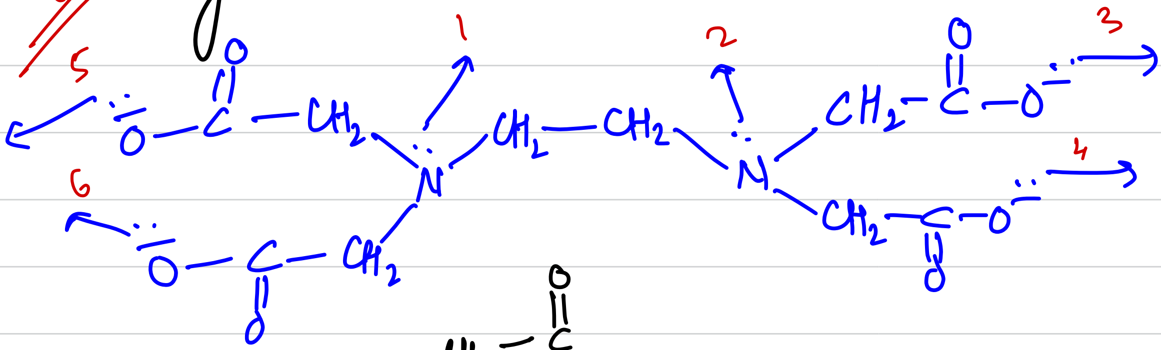
(b) Diethylenetriamine (Dien):-

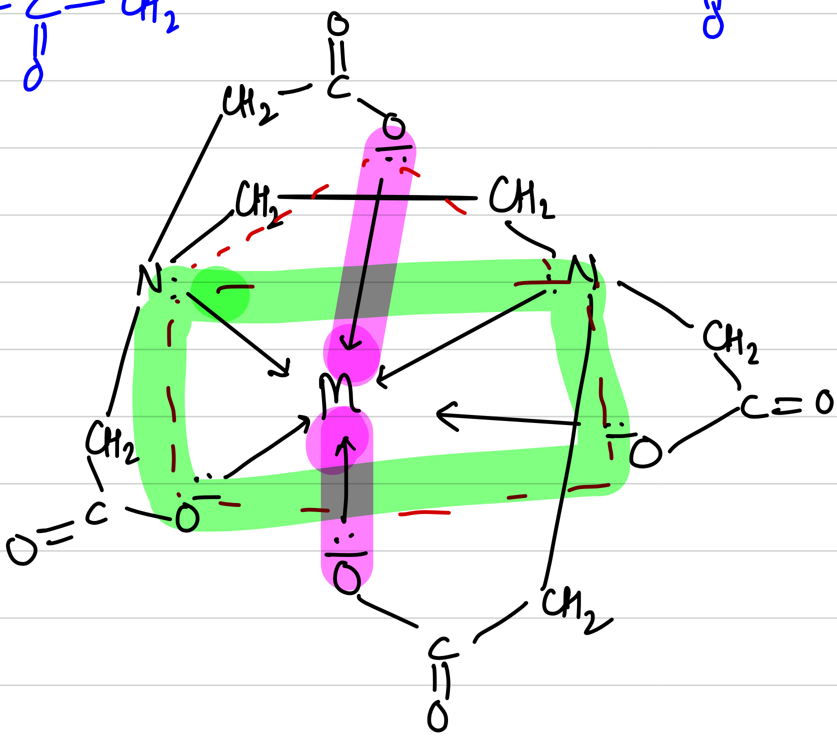


(4) hexadentate ligand (EDTA):-

Can donate 6 e^- pairs at a time.

ethylenediaminetetraacetate





Ambidentate ligand:-

Those ligands which have two different atom for donation of e^-s but can donate only one e^- pair at a time.

eg:

$\leftarrow :\ddot{C}N$ / $\leftarrow :N \equiv C^{\ominus}$

$\leftarrow :\ddot{O}-N=O$ / $\leftarrow :N(=O)-O^-$

$\leftarrow :\ddot{S}-C\equiv N$ / $\leftarrow :\ddot{N}=C=S$

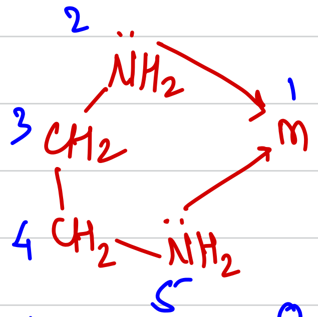
Chelating ligands:-

Chelation → ring formation.

* Polydentate ligand/ bidentate ligands where structure permits the attachment of two or more donor sites to attach to same metal ion simultaneously, thus closing one or more rings are called chelating ligands.

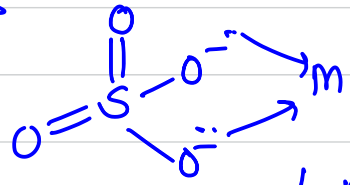
eg: (en).

en →



5-membered ring

SO_4^{2-}



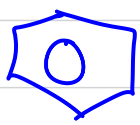
(4-membered ring) (due to the formation of 4-membered ring this does not act as a bidentate Ligand).

Classification of ligands on the basis of charge :-

① Negatively charged ligands :-

eg: F^-, Cl^-, Br^-, I^-, OH^-, CN^-, SH^-, O^{2-} (oxide), O_2^- (Superoxide), O_2^{2-} (peroxide), CN^-, SCN^-, etc.

② Neutral ligands :-

eg: NH_3, H_2O, en, dien, PH_3 , CO, PF_3.

③ Positively charged ligands :-

NO^+ → nitrosonium ion

$H_2N-NH_3^+$ → hydrazinium ion

③ On the basis of donor & acceptor ability :-

(A) Ligands which contain vacent π-orbital (type) that can recieve back donation from the metal ion.

eg: CO, NO, $\ddot{\bar{C}}N$, $N\equiv\ddot{\bar{C}}$, $R_3P:$ → vacant d-orbital

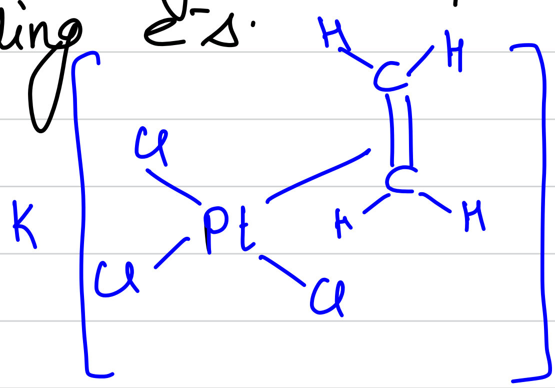
$-\dot{N}=O$ $\ddot{\bar{C}}\equiv N$

$\ddot{M} \leftarrow \bar{C}\equiv \overset{\oplus}{O}$ → CO is a sigma donor & π-acceptor.

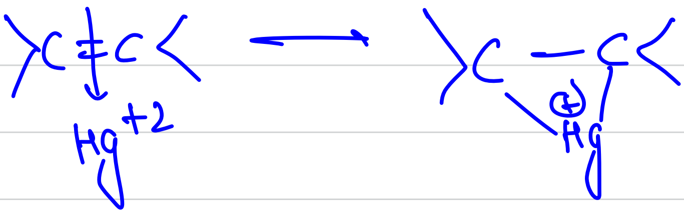
ⓑ Ligands which donot have vacent orbital to recieve back donation.

$M \leftarrow \ddot{F}^{-}$

ⓒ Ligands having no lone pair of e^- but having π-bonding e^-s.



Zises salt



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Co-ordination no :-

$$CN = \sum_{i=1}^{n} (\text{no of ligands} \times \text{ligancy})$$

$K_4[\overset{+2}{Fe}(CN)_6]$ $\quad CN = 6 \times 1 = \underline{\underline{6}}$

$1 \times 4 + x + (-1) \times 6 = 0 \quad x = 2$

$[\overset{+3}{Co}(en)_3]^{3+}$ $\quad CN = 3 \times 2 = 6$

$x + (0) \times 3 = +3$

$x = \underline{\underline{+3}}$

$[PtCl_4]^{2-}$ $\quad CN = 4 \times 1 = \underline{\underline{4}}$.

$x + (-1) \times 4 = -2$

$x = \underline{\underline{+2}}$

Oxidation number :- / state

It is the no which represents the electric charge on the central metal atom of a complex ion.

Q. Find out the CN & ON of the following :-

① $K_4[\overset{+2}{Fe}(CN)_6]$ $CN = 6.$

Potassium hexacyanoferrate (II)

② $[\overset{+2}{Cu}(NH_3)_4]SO_4$ $CN = 4 \times 1 = 4.$

$x + (0) \times 4 + (-2) = 0$

$x = \underline{\underline{2}}$

tetraammine copper(II) sulphate.

③ $[\overset{0}{Ni}(CO)_4]$ $CN = 4 \times 1 = 4$ tetracarbonylnickel (0)

$x + (0) \times 4 = 0$

$x = \underline{\underline{0}}$

④ $K_2[\overset{+2}{Ni}(CN)_4]$ $CN = 4 \times (1) = 4$ Potassium tetracyanonickelate (II)

$1 \times 2 + x + (-1) \times 4 = 0$

$x = \underline{\underline{+2}}$

⑤ $K_2[\overset{+2}{Hg}I_4]$ $CN = 4 \times 1 = 4$

$1 \times 2 + x + (-1) \times 4 = 0$

$x = \underline{\underline{+2}}$

Potassium tetraiodidomercurate (II).

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Nomenclature of co-ordination compounds:-

Rules

① In common salts cation is named first & then the anion.

② In the complex ion (cationic or anionic) ligands are named first followed by the name of CMI.

③ The ON of CMI is indicated by Roman numerals in the ( ) after the name of CMI. The zero oxidation state is indicated by (0).

④ In case of neutral complexes, it is named as one word.

$[Ni(CO)_4]$ tetracarbonylnickel(0).

⑤ There should be space b/w the name of cation & the anion.

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⑥ If the co-ordination sphere is anionic, then the name of CMI is written by adding a suffix -ate in the name of metal.

iron → Ferrate

Zn → Zincate

Al → Aluminate

Au → Aurate

Hg → mercurate

Ag → Argentate

Cr → Chromate

⑦ If the co-ordination sphere is positively charged then the name of the CMI is written as such.

Nomenclature of ligands:-

(a) If there are two or more different kinds of ligands are present then they are named in alphabatical order.

(b) If a particular ligand is more than one times in the complex, the no. is indicated by

di → 2
tri → 3
tetra → 4
penta → 5
hexa → 6

(c) If the name of ligand contains (bi, -tri, -tetra then

bis - 2
tris → 3
tetrakis → 4
pentakis → 5

$[\overset{+2}{Zn}(en)_2]^{2+}$ → bis-(ethylenediamine)zinc(II) ion

Nomenclature of anionic ligands:-

In general the name of ligand is written by adding a suffix $\boxed{\text{'o'}}$ & removing $\boxed{\text{'e'}}$.

$H^{\ominus}$ → hydride → hydrido

O^{2-} → oxide → oxo

O_2^{2-} → peroxide → peroxo

$\overset{-}{O}H$ → hydroxide → hydroxido

F^- → fluoride → fluorido or fluoro

Cl^- → Chloride → Chloro or chlorido

Br^- → Bromide → Bromo or Bromido

I^- → Iodide → Iodido or Iodo

$\overset{-}{C}N$ → cyanide → cyanido or cyano.

NO_3^- → nitrate → nitrato

$NH_2-CH_2-COO^-$ → glycinate → glycinato

SO_4^{2-} → sulphate → sulphato

Nomenclature of neutral ligands:-

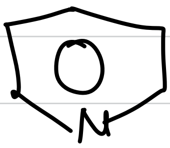
In general neutral ligands name do not change with some special name.

$H_2O \rightarrow$ aqua

$NH_3 \rightarrow$ ammine

$CO \rightarrow$ carbonyl

$NO \rightarrow$ nitrosyl


$\rightarrow$ pyridine

$CH_3-NH_2 \rightarrow$ methyl amine

$O_2 \rightarrow$ dioxygen

$PH_3 \rightarrow$ phosphene.

$$\begin{matrix} CH_2-NH_2 \\ | \\ CH_2-NH_2 \end{matrix}$$

ethylenediamine

$$NH_2-\overset{O}{\overset{\parallel}{C}}-NH_2 \rightarrow \text{urea}$$

Werner's Theory :-

Metals possesses two types of valencies.

Primary valency → Ionizable → outside the co-ordination sphere

Secondary valency → Non-ionizable. → Inside the coordination sphere.

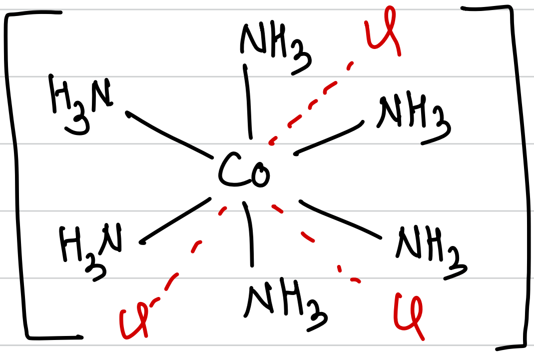
$CoCl_3 \cdot 6NH_3 + AgNO_3 \longrightarrow AgCl\downarrow$ white ppt.

$[Co(NH_3)_6]Cl_3 + AgNO_3 \longrightarrow 3AgCl\downarrow$
[3 ions Cl^-]   excess

$[Co(NH_3)_5Cl]Cl_2 \cdot NH_3 + AgNO_3 \longrightarrow 2AgCl\downarrow$
(2 Cl^- ions)

$[Co(NH_3)_4Cl_2]Cl \cdot NH_3 + AgNO_3 \longrightarrow AgCl\downarrow$
(1 Cl^- ion)

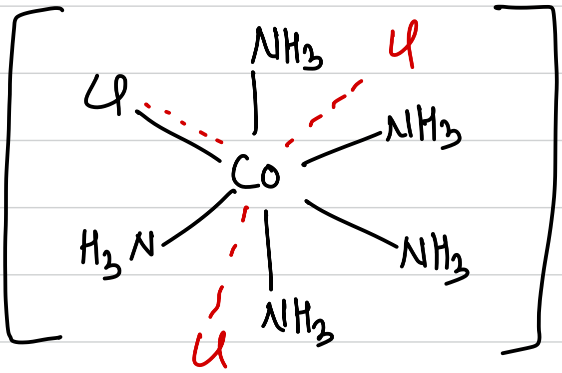
$[CoCl_3(NH_3)_3] \cdot 3NH_3 \rightarrow$ this complex won't form.



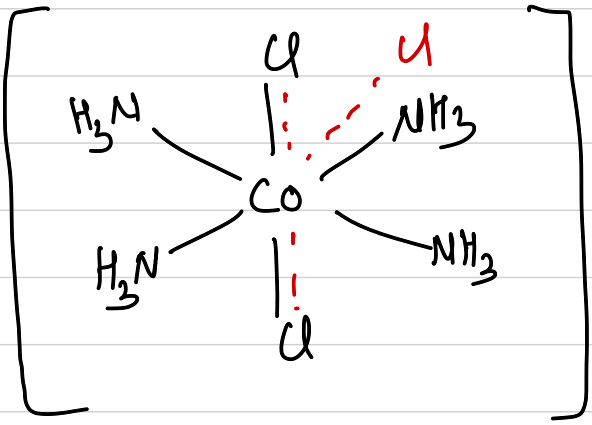
- - - - primary valency

——— secondary valency

(yellow colour)



(purple colour)



(green/violet)

Bonding Theories in Co-ordination Complexes

1. VBT → Valence Bond theory
2. CFT → Crystal Field theory
3. LFT → Ligand Field theory → not in syllabus.

Valence Bond theory :-

This theory tells us about the hybridization of the complex as well as the shape of the complex.

Coordination No.	hybridization	Shape.
2	sp	Linear
3	sp^2	trigonal planar
4	sp^3	tetrahedral
	d^3s	tetrahedral
	dsp^2	square planar
5	$\left.\begin{matrix} dsp^3 \\ sp^3d \end{matrix}\right\}$ →	trigonal bipyramidal
6	$\left.\begin{matrix} sp^3d^2 \\ d^2sp^3 \end{matrix}\right\}$ →	octahedral

hybridization	orbitals Used
sp	s & $p_{x/y/z}$.
sp^2	s & $p_x p_y$ / $p_y p_z$ / $p_z p_x$.
sp^3	s & p_x p_y p_z.
d^3s	d_{xy}, d_{yz}, d_{zx} & s.
dsp^2	$d_{x^2-y^2}$, s & p_x p_y
sp^3d^2	s $p_x p_y p_z$, $d_{x^2-y^2}$, d_{z^2}.
d^2sp^3	$d_{x^2-y^2}$ d_{z^2}, s p_x p_y p_z.
sp^3d^3	s p_x p_y p_z $d_{x^2-y^2}$, d_{z^2}, d_{xy}

Valence Bond Theory for co-ordination no 6 :-

Strength of donation of e^-s.

C-donors > N-donors > O-donor > halide-donor.

$$\underbrace{\ddot{C}N^-,\ CO > NH_3,\ en}_{\text{(strong ligands)}} > \underbrace{C_2O_4^{2-},\ \ddot{O}H^- > F^-,\ Cl^-,\ Br^-,\ I^-}_{\text{(weak ligand)}}$$

⊛ $[\overset{+2}{Fe}Cl_6]^{4-}$ $\qquad$ Cl^- is a weak field ligand.

$_{26}Fe \rightarrow [Ar]\ 3d^6, 4s^2$

$_{26}Fe^{+2} \rightarrow [Ar]\ 3d^6$

3d					4s	4p			4d				
↑↓	↑	↑	↑	↑	xx	xx	xx	xx	xx	xx			

$$\underbrace{4s\ 4p\ 4d}_{sp^3d^2}$$

hybridization $\Rightarrow sp^3d^2$

shape → octahedral

magnetic moment $\mu = \sqrt{n(n+2)}$ BM $\quad$ $n \Rightarrow$ no of unpaired e^-s.

$\mu = \sqrt{4(4+2)}$ BM $= \sqrt{24}$ BM.

paramagnetic in nature.

high spin complex. | outer orbital complex.

(*) $[\overset{+2}{Fe}(CN)_6]^{4-}$

$\ddot{C}N^{-} \rightarrow$ strong field ligand.
↓
will cause pairing

$_{26}Fe \rightarrow [Ar]\, 4s^2, 3d^6$

$_{26}Fe^{+2} \Rightarrow [Ar]\, 3d^6$

3d: | ↑↓ | ↑ | ↑ | ↑ | ↑ | 4s: | × × | 4p: | ×× | ×× | ×× | 4d: | | | | |

| ↑↓ | ↑↓ | ↑↓ | ×× | ×× |

$\underline{\underline{d^2sp^3}}$

hybridization $\Rightarrow d^2sp^3$ shape → octahedral

$\mu = \sqrt{n(n+2)}$ BM $= 0$ BM

since. $\underline{\underline{n=0}}$.

diamagnetic in nature.

Low spin complex. / inner orbital complex.

③ $[Cr(H_2O)_6]^{3+}$

CN = 6
ON = 3

$_{24}Cr \Rightarrow [Ar]\, 4s^1\, 3d^5$

$_{24}Cr^{+3} \Rightarrow [Ar]\, 3d^3$

H_2O is a weak field ligand.

3d					4s	4p			4d				
↑	↑	↑	xx	xx	xx	xx	xx	xx					

$\underbrace{\text{xx xx xx xx xx xx}}_{d^2sp^3}$

shape → octahedral

$\mu = \sqrt{3(3+2)} = \sqrt{15}$ BM

paramagnetic / inner orbital complex.

(4) $\overset{+3}{[Cr(CN)_6]^{3-}}$ $\ddot{C}N$ strong ligand.

3d					4s	4p			4d				
↑	↑	↑	xx	xx	xx	xx	xx	xx					

$\underbrace{\text{xx xx xx xx xx xx}}_{d^2sp^3}$

shape → octahedral

$\mu = \sqrt{3(3+2)} = \sqrt{15}$ BM

paramagnetic / inner orbital complex. ✓

Imp (*) If the electronic configuration of central metal ion is d^1, d^2 or d^3 in case co-ordination no. 6. Then hybridization will always be d^2sp^3 irrespective of the ligand.

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(5) $[Fe(CN)_6]^{3-}$

CN = 6
ON = 3

$\ddot{C}N$ is a strong ligand.

$_{26}Fe^{+3} \Rightarrow [Ar]\ 3d^5$

↑	↑	↑	↑	↑				

↑↓	↑↓	↑	xx	xx	xx	xx	xx	xx

$d^2sp^3 \rightarrow$ octahedral

$\mu = \sqrt{1(1+2)}\ BM = \sqrt{3}\ BM.$

paramagnetic. / low spin complex inner orbital complex.

(6) $[Ni(H_2O)_6]^{2+}$

CN = 6
ON = 2

H_2O is a weak ligand.

$28\ Ni \rightarrow [Ar]\ 4s^2 3d^8$

$28\ Ni^{+2} \Rightarrow [Ar]\ 3d^8$

3d					4s	4p			4d				
↑↓	↑↓	↑↓	↑	↑	xx	xx	xx	xx	xx	xx			

sp^3d^2

paramagnetic

$\mu = \sqrt{8}\ BM.$

outer orbital complex.

7. $[Ni(NH_3)_6]^{+2}$ $CN = 6$, $ON = 2$

$_{28}Ni^{+2} = [Ar]3d^8$

NH_3 is a strong Ligand.

3d	4s	4p	4d
↑↓ ↑↓ ↑↓ ↑ ↑	xx	xx xx xx	xx xx _ _ _

$\mu = \sqrt{8}$ BM.

paramagnetic.

$sp^3d^2 \rightarrow$ outer orbital complex.

$\boxed{d^4, d^5, d^6, d^7}$ × → check for strong/weak Ligand & pairing.

(*) In case d^8, d^9, d^{10} systems having $CN = 6$.
hybridization will be sp^3d^2, irrespective of Ligand.

⊛ VBT for Co-ordination no. 4 :-

① $[NiCl_4]^{2-}$ → CN = 4, ON = 2 $\qquad$ Cl^- weak ligand.

$_{28}Ni^{+2} \Rightarrow [Ar]\,3d^8$

3d					4s	4p		
↑↓	↑↓	↑↓	↑	↑	xx	xx	xx	xx

sp^3 → tetrahedral

$\mu = \sqrt{2(2+2)} = \sqrt{8}$ BM, paramagnetic complex.

② $[Ni(CN)_4]^{2-}$ → CN = 4, ON = 2 $\qquad$ CN^- → strong ligand.

$_{28}Ni^{+2} \Rightarrow [Ar]\,3d^8$

3d					4s	4p		
↑↓	↑↓	↑↓	↑	↑				

3d					4s	4p		
↑↓	↑↓	↑↓	↑↓	xx	xx	xx	xx	

dsp^2 → square planar

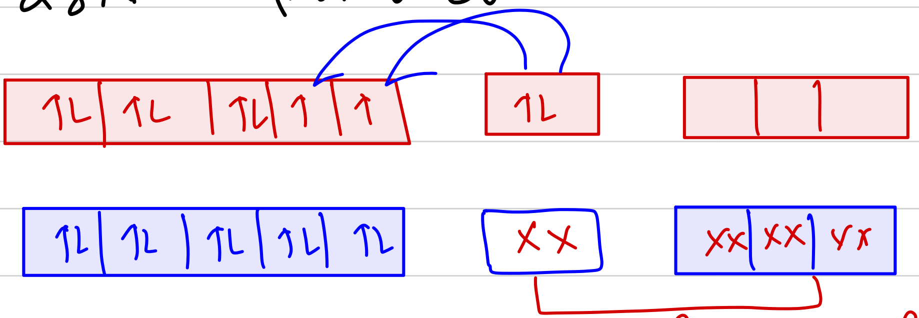
diamagnetic complex.

$\underline{\mu = 0}$.

(3) $[\overset{0}{Ni}(CO)_4]$ CN = 4, ON = 0.

CO → strong donor.

$_{28}Ni \Rightarrow [Ar]\, 4s^2\, 3d^8$



sp^3 → tetrahedral

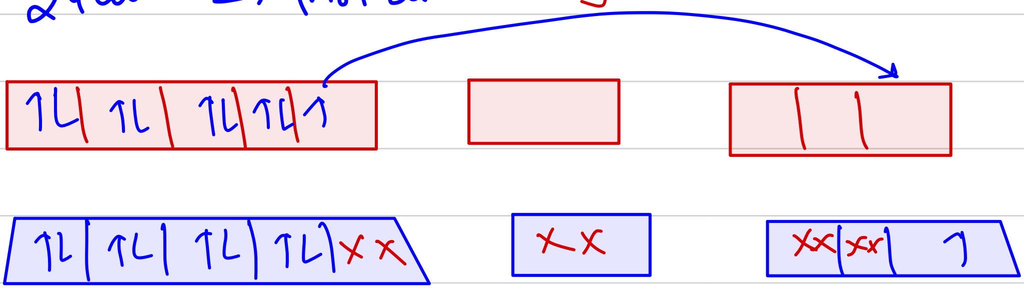
$\mu = 0$, diamagnetic.

(4) $[Cu(NH_3)_4]^{+2}$ CN = 4, ON = 2

NH_3 is a strong field ligand.

$_{29}Cu \rightarrow [Ar]\, 4s^1\, 3d^{10}$

$_{29}Cu^{+2} \Rightarrow [Ar]\, 3d^9$



dsp^2 → square planar.

$\mu = \sqrt{1(1+2)} = \sqrt{3}$ BM.

Transference :-

It is the shifting or transfer of an electron from lower energy level to a higher energy level in a complex compound when certain conditions are satisfied is called transference.

1. Only one electron can be transferred.
2. Presence of strong field ligand should be there.
3. There should be possibility of formation of inner orbital complex.

Drawbacks of VBT:-

① It can not explain why some complexes of metal ion are inner orbital where as some other are outer orbital.

② It does not explain variation of magnetic properties of complexes wi temperature.

③ VBT was unable to explain spectral properties of compounds.

④ VBT was unable to explain the stability of complexes.

CRYSTAL FIELD THEORY

According to this theory, Ligands & metal ions both are treated as point charges. & there will be force of attraction between the central metal ion & the ligand.

There is no orbital interaction between the metal ion & the ligand.

⊛ The d-orbitals of the metal have same energy (degenerate) in free atom. However, when a complex is formed the ligands destroy the degeneracy of these orbitals.

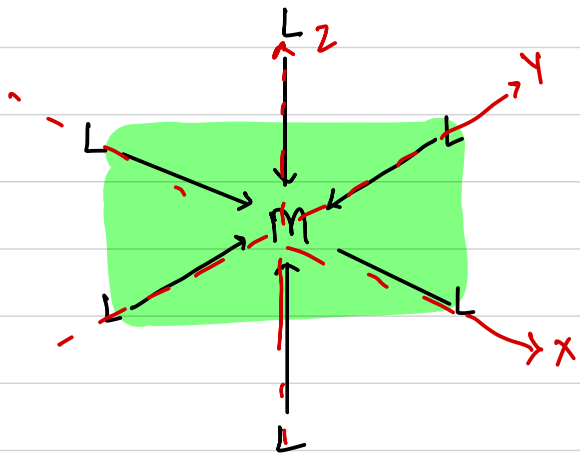
d-orbitals are of two types

- along the axis orbitals
  $d_{z^2}, d_{x^2-y^2}$
- inbetween the axis orbitals.
  d_{xy}, d_{yz}, d_{zx}

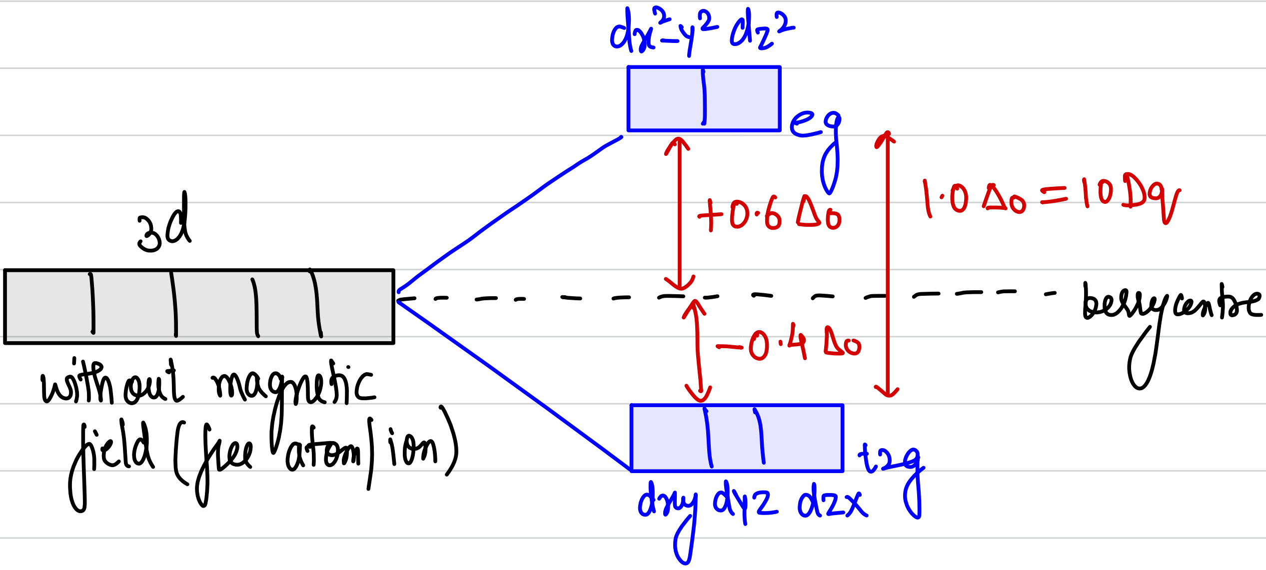
CFT

- Octahedral complex
- Tetrahedral complexes
- Square planar complex.

CFT for octahedral complexes:-

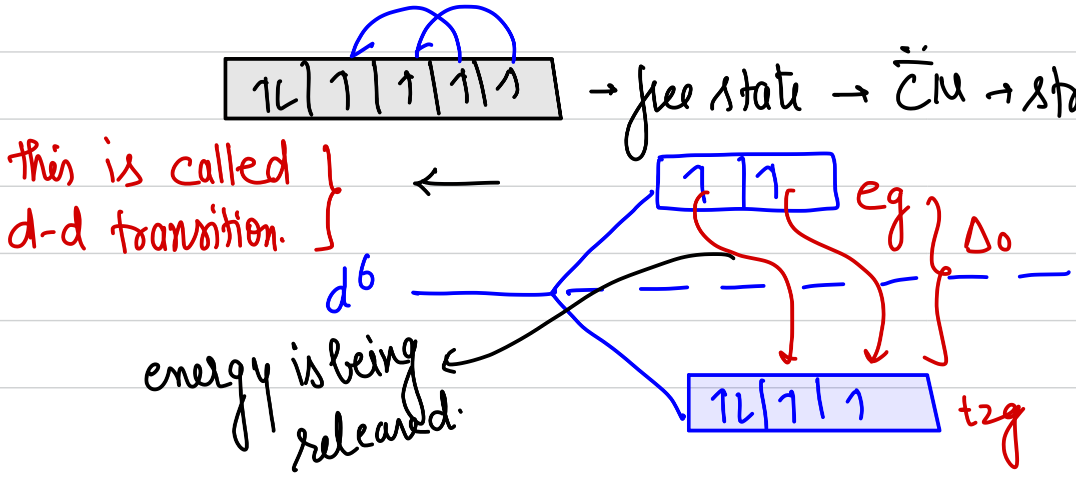


* All the ligands are approaching along the axis. Moving ligands will produce magnetic field. & those orbitals (d) of metal which lie along the axis will get more repelled by the ligand e^-s.



g → garade → centre of symmetry

$[\overset{+2}{Fe}(CN)_6]^{4-} \Rightarrow Fe^{+2} \Rightarrow [Ar]\,3d^6$



$\Rightarrow t_{2g}^4\, e_g^2$

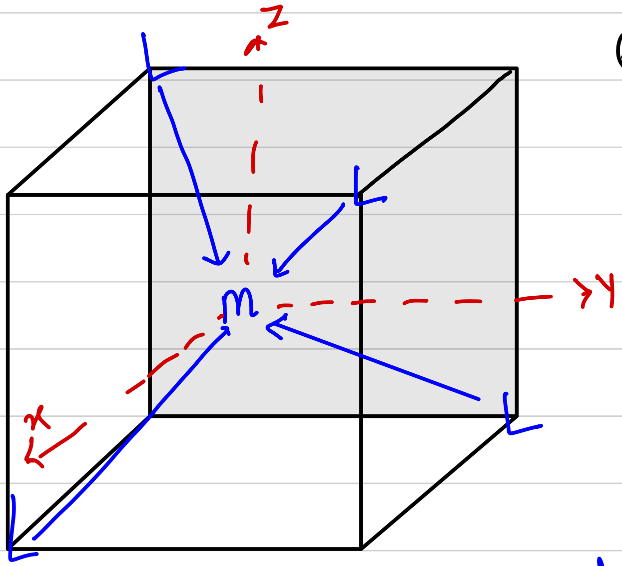
↓ after pairing

$= t_{2g}^6$

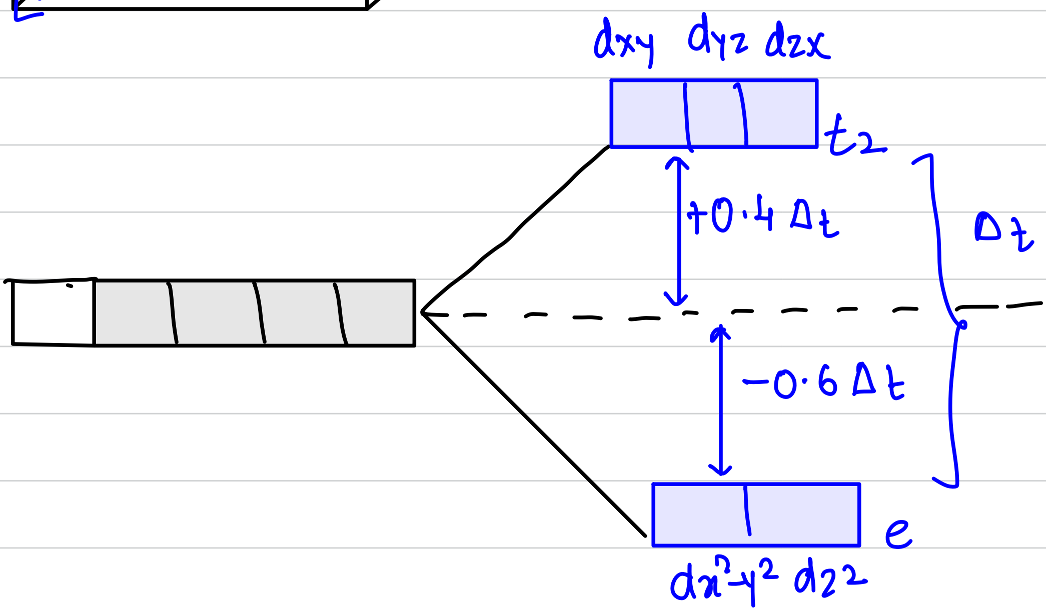
P → Pairing energy.

* ① If $\Delta_o > P$ ⇒ then only pairing will occur.
  ② If $\Delta_o < P$ ⇒ then pairing will not occur.

CFT for Tetrahedral Complexes:-



⊛ In tetrahedral field the ligands are approaching in between the axis. Therefore these d-orbitals of the metal which are in between the axis will get repelled more.

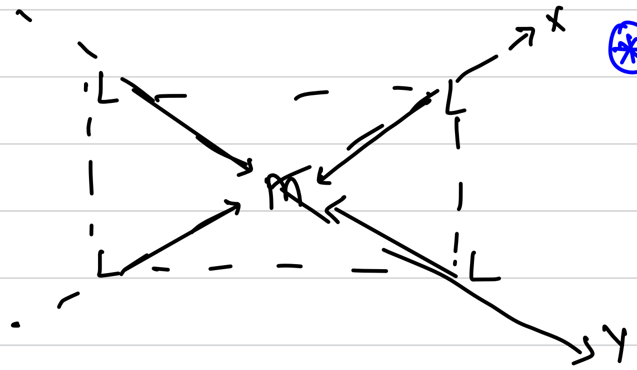


$$\boxed{\Delta_t = \frac{4}{9}\Delta_o}$$

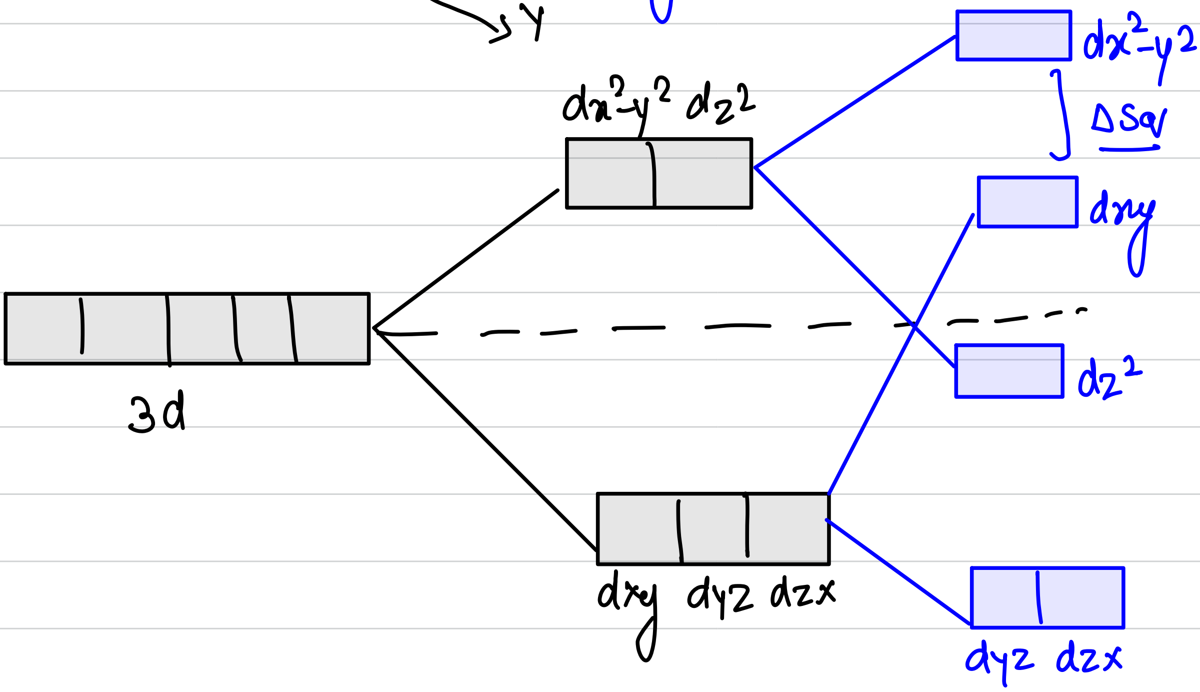
$\frac{2}{3}$ ⇒ because of no of ligands

$\frac{2}{3}$ ⇒ because of direction.

CFT for Square planar complexes :-



✱ In square planar field the four ligands are approaching along the internuclear axis only in x-y plane.



Spectrochemical Series :-

weak field →

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} <$
$< H_2O < NCS^- < EDTA^{4-} < NH_3 < en < CN^- < CO.$

strong field

The spectrochemical series is an experimentally determined series.

It is difficult to explain the order as it incorporates both the effect of σ & π - Bonding

Imp

$$\boxed{\text{halide donor} < \text{O-donors} < \text{N-donor} < \text{C-donor}}$$

How to Calculate the CFSE:-

Crystal field stablization energy:-

for CN=6.

$$\text{CFSE} = (\text{no of } e^-\text{s in } t_{2g} \text{ orbitals} \times (-0.4\Delta_0)) + (\text{no of } e^-\text{s in } e_g \text{ orbital} \times (0.6\Delta_0)) + xP$$

where x is the no of new e^- pair formed due to $\Delta_0 > P$.

for d^6 system:-

Weak Field Ligand	Strong field Ligand.
$d^6 \rightarrow t_{2g}^4 e_g^2$	$d^6 \rightarrow t_{2g}^6 e_g^0$ (due to pairing)
$\text{CFSE} = (4\times-(0.4)\Delta_0) + 2\times(0.6\Delta_0)$	$\text{CFSE} = 6\times-0.4\Delta_0 + 0\times0.6\Delta_0 + 2P$
$= -1.6\Delta_0 + 1.2\Delta_0$	$= -2.4\Delta_0 + 2P$
$= -0.4\Delta_0$	

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Calculation of CFSE for tetrahedral complex :-

$$\boxed{\text{CFSE} = (\text{no of } e^-\text{s in } e \text{ orbital} \times (-0.6\Delta_t)) + (\text{no of } e^- \text{ in } t_2 \text{ orbitals} \times (0.4\Delta_t))}$$

for tetrahedral complexes

$\boxed{\Delta_t < P}$ that's why pairing will never occur in tetrahedral complexes

Factors affecting the magnitude of orbital splitting energy (Δ):-

① Oxidation state of metal ion:-

The magnitude of the splitting energy increases with increase in oxidation state of central metal ion.

$$\left[\overset{+3}{Co}(NH_3)_6\right]^{3+} > \left[\overset{+2}{Co}(NH_3)_6\right]^{2+}$$

$$\left[\overset{+3}{Fe}(CN)_6\right]^{3-} > \left[\overset{+2}{Fe}(CN)_6\right]^{4-}$$

② Nature of ligand:-

Strong field ligands causes greater splitting as compared to weak field ligand.

$$\left[\overset{+3}{Co}F_6\right]^{3-} < \left[\overset{+3}{Co}(H_2O)_6\right]^{3+} < \left[\overset{+3}{Co}(NH_3)_6\right]^{3+}$$

③ Presence of Chelating ligands:-

The chelating ligands or ring formin ligands causes a greater splitting of d-orbitals regardless of donor atoms.

$$[Co(en)_3]^{3+} > [Co(NH_3)_6]^{3+}$$

causes chelation   no chelation

Imp ④ Identity of central metal ion :-

The magnitude of CFSE depends upon the identity of CMI. [i.e. to which particular transition series the metal belongs to].

* The splitting energy obtained in case of 5d-series is 25% more than 4d-series which intern 50% greater than 3d series.

$[NiCl_4]^{2-}$ → 3d → sp^3

$[PdCl_4]^{2-}$ → 4d → dsp^2

$[PtCl_4]^{2-}$ → 5d → dsp^2

$\underbrace{dsp^2 \quad dsp^2}_{\text{square planar}}$

Stability of Complexes:-

In general complexes are more stable than common salts.

But sometimes they dissociate in aq. solution to a certain extent which may vary from complex to complex.

$$[Co(NH_3)_6]^{3+} \rightleftharpoons Co^{+3}_{(aq)} + 6NH_3(aq)$$

$$K_{eq} = 6 \times 10^{-35}.$$

Generally we check for formation constant: (K_f)

Greater the value of K_f greater will be the stability.

⊛ Factors affecting stability of complexes:-

(a) CFSE:-

Greater the value of CFSE greater will be the stability of complex.

(b) Charge density of central metal ion:-

Greater the charge density of CMI greater will be the force of attraction & hence greater will be the stability.

order of stability:- $Cu^{+2} > Ni^{+2} > Co^{+2} > Fe^{+2} > Mn^{+2}$.

DexterChem

(c) Nature of ligand:- Greater the field strength of ligand greater will be the stability

(d) Chelate effect:- Greater the extent of chelation greater will be the stability

(e) EAN Rule:- Effective Atomic Number:

In complexes the CMI tends to attain the nearest noble gas configuration by accepting the e^- or by donating the e^-.

$$\boxed{EAN = Z - ON + 2 \times CN}$$

Z → atomic no.
ON → oxidation no.
CN → co-ordination no.

$\overset{+2}{[Fe(CN)_6]}^{4-} \Rightarrow EAN = 26 - 2 + 2 \times 6$
$= \underline{\underline{36}}$

$\overset{+3}{[Co(NH_3)_6]}^{3+} \Rightarrow EAN = ?$
$= 27 - 3 + 2 \times 6 = \underline{\underline{36}}$ ✓

$\overset{0}{[Cr(CO)_x]} \Rightarrow EAN = 24 - 0 + 2 \times x = 36$
$x = \underline{\underline{6}}$ ✓

$[\overset{0}{V}(CO)_6] \Rightarrow EAN = 23 - 0 + 2\times 6 = \underline{\underline{35}} \longrightarrow 36]$ reduction

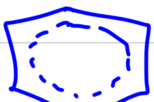
$[\overset{0}{Cr}(CO)_6] \Rightarrow EAN = 24 - 0 + 2\times 6 = 36 \rightarrow$ (neither oxidation nor reduction)

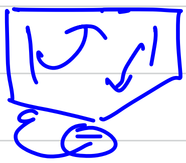
$[\overset{0}{Mn}(CO)_6] \Rightarrow EAN = 25 - 0 + 2\times 6 = 37 \rightarrow 36$

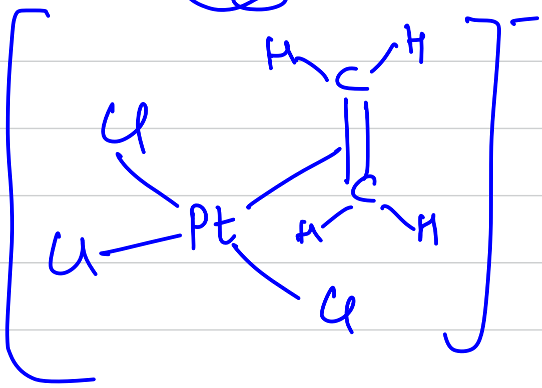
$\hookrightarrow$ oxidation.

Hapticity :- (η) :-

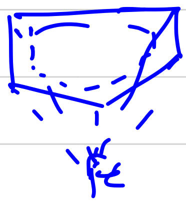
The total no of atoms in the ligand to which the donatable πe^- cloud belongs to is called hapticity.

 $\Rightarrow$ π-e^- belongs to 6 carbons - η^6.

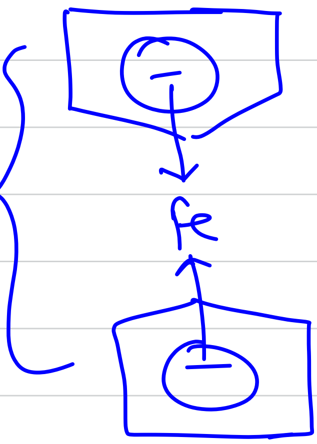
 $\equiv$  $\Rightarrow \eta^5$



$\Rightarrow [Pt(\eta^2\text{-}C_2H_4)Cl_3]^-$



Sandwich



Ferrocene $\rightarrow [\overset{+2}{Fe}(\underline{\eta^5\text{-}C_5H_5})_2]$

$EAN = 26 - 2 + 2\times 6$

$= \underline{\underline{36}}$

⊛ Colours in Compounds:-

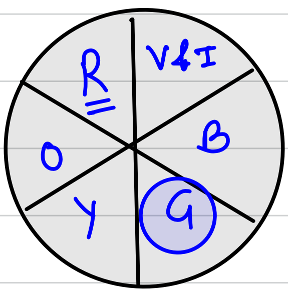
Reasons for colour

1. d-d transitions
2. Ligand to metal charge transfer (LMCT)
3. Resonance (delocalization of charge) → colours in dye & indicators)
4. Polarization
5. F-centres → metal excess defect
6. HOMO-LUMO transition.

1. d-d transitions:-

The transition d-e^- from higher energy orbitals to lower energy d-orbitals. When the transition occurs or e^- jumps from higher level to lower level it will emit some energy & correspondingly some light of some wavelength in visible region.

colour wheel

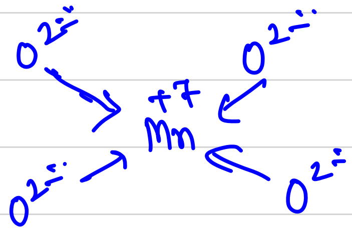


If a compound has absorbed the green light it will emit the red light.

Imp ✱ If CFSE is more the wavelength will be less of absorbed radiation & higher will be the wavelength of emitted radiation.

② **LMCT** :- complexes having d^0 configuration:-

$\overset{+7}{MnO_4^-}$ $\quad Mn^{+7} = 3d^0$ → Pink or purple colour is due to $O^{2-} \rightarrow Mn$ e^- transfer.



→ Td. → d^3s.

✱ $K_2\overset{+6}{Cr_2}O_7$ → LMCT → orange

$\overset{+6}{Cr}O_5$ → LMCT → blue

④ Colours due to polarization:-

$4d^{10}$. $\quad \overset{+1}{Ag}Cl \quad \overset{+}{Ag}Br \quad \overset{+1}{Ag}I$

white $\quad$ Pale yellow $\quad$ Bright yellow

⑤ **F centres**:-

$Zn_{1.010}$

$$\underset{\text{white}}{ZnO} \xrightarrow{\Delta} \underset{\text{yellow}}{ZnO} \Rightarrow (Zn_{1+x}O)$$

(metal excess defect)

$$\underset{\text{Yellow}}{PbO} \xrightarrow{\Delta} \underset{\text{Red}}{PbO}$$

⑥ Homo LUMO transition → in halogen.

↳ F_2 → colourless $\quad Cl_2$ → greenish yellow $\quad Br_2$ → Reddish Brown

I_2 → violet

Isomerism in Co-ordination Compounds

- Structural
  - Ionization
  - Hydrate
  - Linkage
  - co-ordination position
  - co-ordination sphere isomerism
- stereo
  - Geometrical
  - optical

Structural Isomerism :-

① Ionization Isomerism :-

The compounds having same emperical formula but producing different ions in aqueous solution are called ionization isomers.

[In this the primary and secondary valencies get interchange.]

$$[Co(NH_3)_5Br]SO_4 \xrightarrow{H_2O} [Co(NH_3)_5Br]^{+2} + SO_4^{-2} \xrightarrow{AgNO_3} Ag_2SO_4 \downarrow \text{ white}$$

$$[Co(NH_3)_5SO_4]Br \xrightarrow{H_2O} [Co(NH_3)_5SO_4]^{+} + Br^{-} \xrightarrow{AgNO_3} AgBr \downarrow \text{ Pale yellow}$$

② Hydrate isomerism :-

The compound having same emperical formula but different no of molecules of H_2O of hydration are called hydrate isomers.

$[Cr(H_2O)_6]Cl_3 \longrightarrow$ violet color.

$[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O \rightarrow$ Light green

$[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O \rightarrow$ dark green

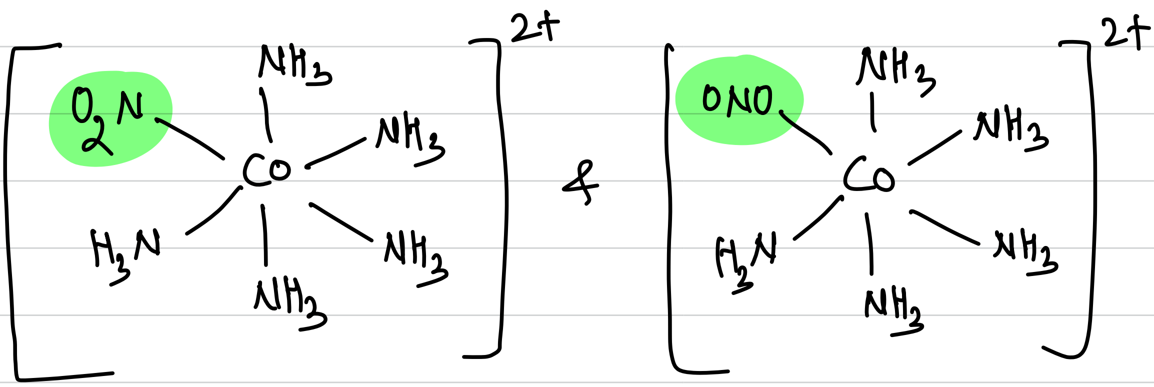
$\underbrace{[Cr(H_2O)_3Cl_3]}_{\text{neutral complex}} \cdot 3H_2O \longrightarrow$ not possible.

neutral complex → water can not be held by neutral complexes.

③ Linkage isomerism :-

Linkage isomerism is shown by ambidentate ligands.

eg: $\bar{C}N$, NO_2^-, $\ddot{S}CN$ etc.



4. Co-ordination sphere Isomerism :-

The compounds which have both cationic & anionic part as complex show this type of isomerism by exchange of the ligands b/w them.

$$[\overset{+3}{Co}(NH_3)_6]^{+3}\ [\overset{+3}{Cr}(CN)_6]^{-3}$$

$$\downarrow$$

$[Co(NH_3)_5CN]\ [Cr(CN)_5NH_3]$ ✓

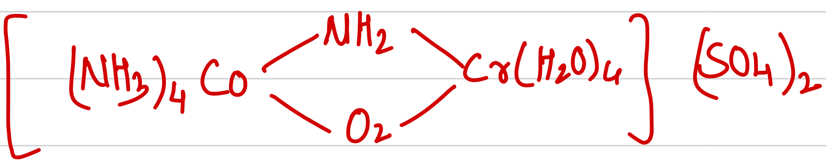
$[\overset{+3}{Co}(CN)_6]^{-3}\ [\overset{+3}{Cr}(NH_3)_6]^{+3}$ ✓

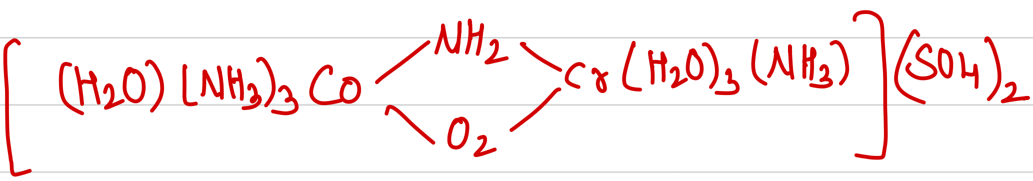
$$\equiv$$

$$[Cr(NH_3)_6]\ [Co(CN)_6]$$

5. Co-ordination position isomerism :-

It occurs when ligands have different positions w.r.t. the two central metal ions in a polynuclear complex.





Geometrical Isomerism:-

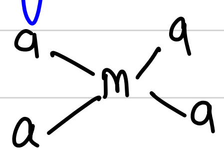
① In case of co-ordination no 4:-

(a) Tetrahedral complexes:-

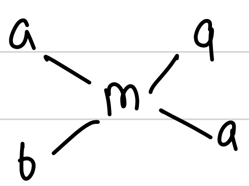
Td complexes do not show Geometrical Isomerism (G.I.) unless the ligand itself show G.I.

(b) Square planar Complexes:-

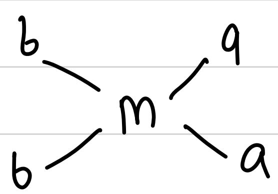
If the angle b/w the ligands is 90° → cis
If the angle b/w ligands is 180° → trans.

Ma_4 → 

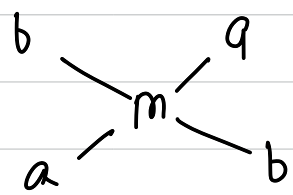
no G.I.

Ma_3b → 

no. G.I.

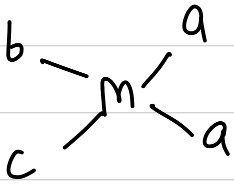
Ma_2b_2 

(a,b) (a,b)
cis form



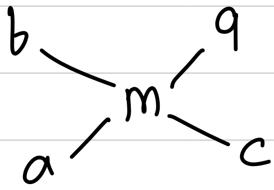
(a,a) (b,b)
trans form.

Ma_2bc :-



cis

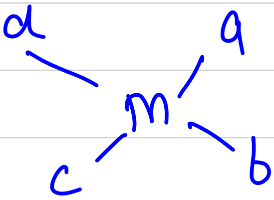
(a,c) (a,b)



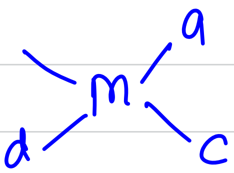
trans

(b,c) (a,a)

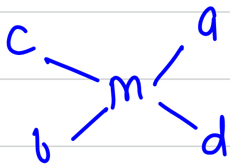
Mabcd



(ac) (bd)

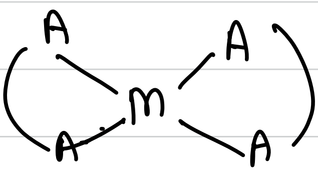


(bc) (ad)

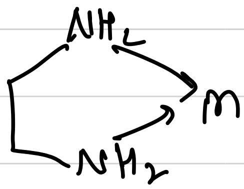


(ab) (cd)

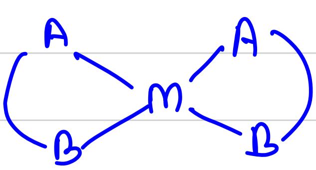
$M(AA)_2 \Rightarrow$ AA → bidentate ligand having same donor atom.



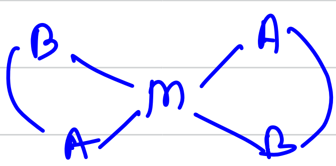
no G.I



$M(AB)_2$



(AB)(AB)

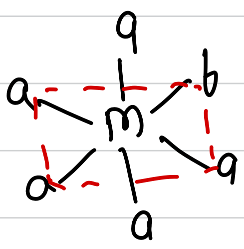


(AA) (BB)

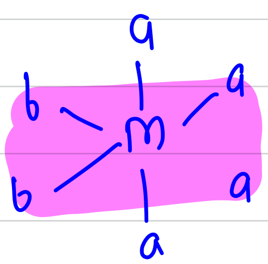
Geometrical Isomerism in case of octahedral complexes:-

$Ma_6 \rightarrow$ no G.I.

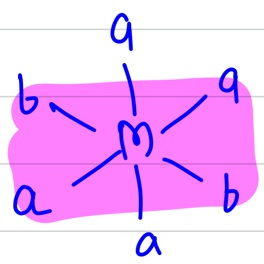
$Ma_5b \rightarrow$ no G.I



Ma_4b_2:-



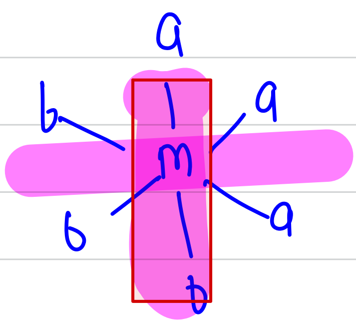
cis



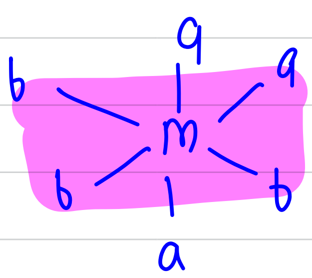
trans

Imp

Ma_3b_3

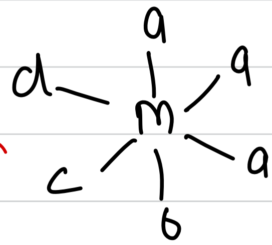
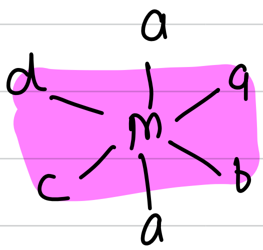


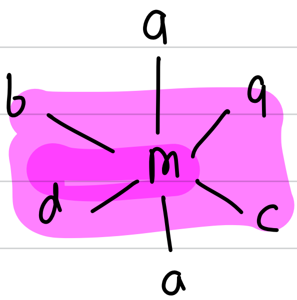
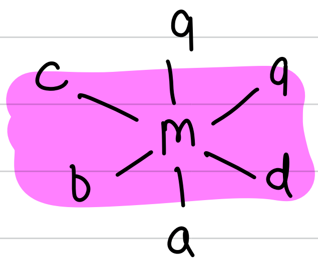
facial isomer (fac)
no. O.I.



meridonial
(mer)

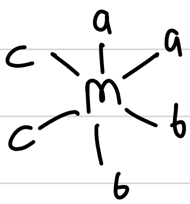
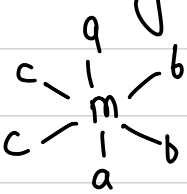
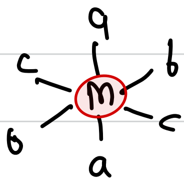
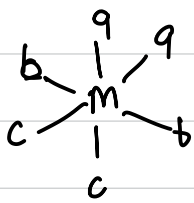
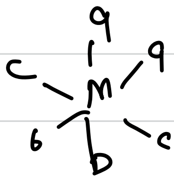
Ma_3bcd :-

(2) x  

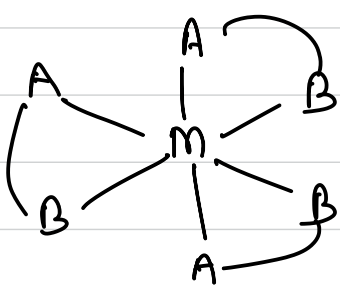
 

HW

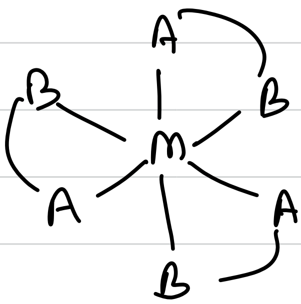
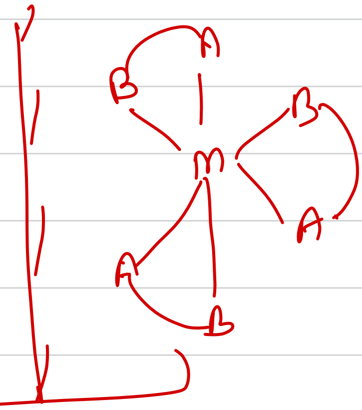
$Ma_2b_2c_2 \Rightarrow$ no of G.I. $\longrightarrow$ 5

$M(AB)_3$

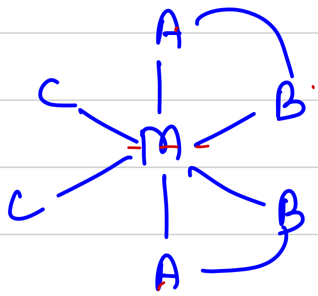


x 2

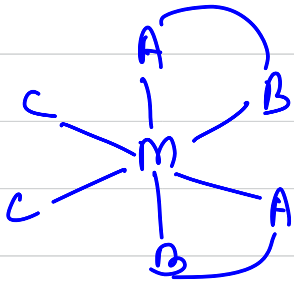
 

non super imposable mirror images.
(enantiomers)

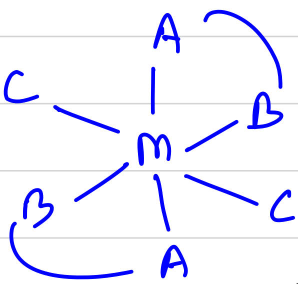
$[M(AB)_2C_2]$



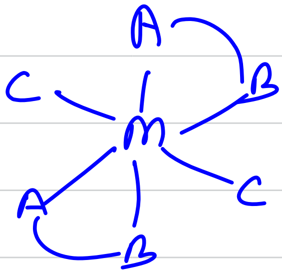
(AA) (BC) (BC)



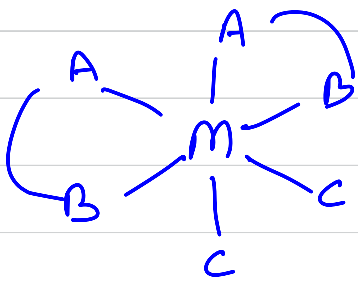
(AB) (AC) (BC)



(AA) (BB)(CC)



AB AB CC



(AC) (AC) (BB)

DexterChem

Optical Isomerism:-

In General square planar complexes do not show O.I.

Tetrahedral complex will show O.I. if all the ligands (4) are different.

Optical isomerism in Oh complexes:-

For a compound to show O.I.

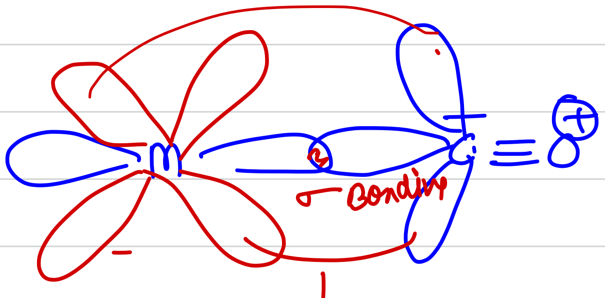
POS should be absent. ✗

COS should be absent. ✗

Metal carbonyls :-

$$M \leftarrow \bar{C} \equiv O^{\oplus}$$

CO is a σ donor & π-acceptor



π-Bonding

CO → 3
CO^+ → 3.5
2σs*

$$\ddot{M} - C \equiv O^{\oplus} \rightarrow M \rightarrow C \rightarrow$$ that bond becomes stronger due to back bonding.

↓

Synergic Bonding

Imp:- * The strength of backbonding can be compared by the following factors.

① Oxidation state of metal ion :-

Lower the oxidation state of CMI greater will be the electron density on the metal & hence greater will be the extent of back bonding.

Greater the extent of back bonding stronger will be M→C bond & weaker will be C-O Bond.

DexterChem

Weaker the C–O bond:-

Bond length of C–O will increase
Bond strength will decrease.

$[\overset{+1}{Mn}(CO)_6]^+$ $[\overset{0}{Mn}(CO)_6]$ $[\overset{-1}{Mn}(CO)_6]^-$

order of Back bonding : 3 > 2 > 1

C–O Bond length :- 3 > 2 > 1

② If the oxidation state of CMI is same then compare Zeff of CMI. Higher the Zeff weaker will be the M→C Back bonding & smaller will be C–O bond length.

$[Ni(CO)_4] < [Co(CO)_4]$ M→C Bond strength.

Calculation of no of M-M bonds in polynuclear complexes.

$(Co)_2(CO)_8 \rightarrow$ no of Co — Co bond

(dicobalt octacarbonyl)

Calculate

* Total valence e^- in the entire molecule. (i.e., the no of valence e^-s of metal + no of e^- from each ligand & charge). $\longrightarrow$ A

* Substract this no from $(n \times 18)$ where n is the no of metal ions in the complex. $\Rightarrow (n \times 18) - A = B \longrightarrow$

* $\frac{B}{2}$ gives the total no of M-M bonds in the complex.

$(Co)_2(CO)_8$ $\quad$ $_{27}Co \Rightarrow [Ar]\, 3d^7, 4s^2$ 9

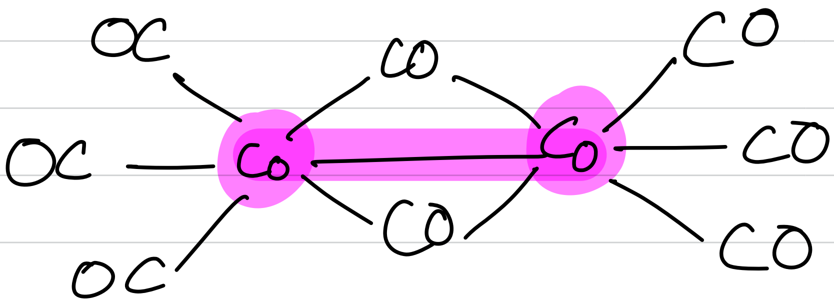
$A = 9 \times 2 + 8 \times 2 = 18 + 16 = 34$

$B = (18 \times n) - A$

$= (18 \times 2) - A = 36 - 34 = \underline{\underline{2}}$

no of Co - Co Bonds $= \frac{2}{2} = 1.$

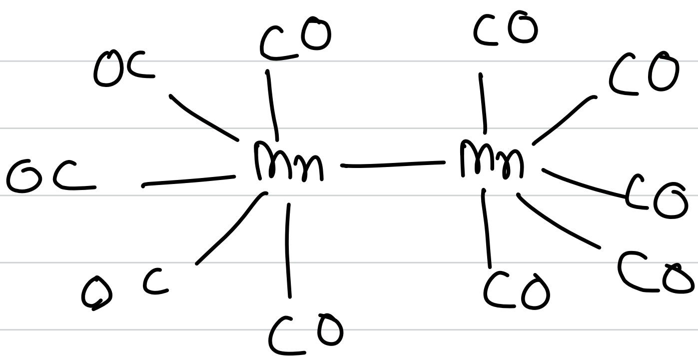
$Mn_2(CO)_{10}$



$(Mn_2(CO)_{10})$

$$A = 7\times2 + 2\times10 = 34$$
$$B = (18\times n) - A$$
$$= 18\times2 - 34 = 2$$

$$\text{no of Bonds} = \frac{2}{2} = 1$$



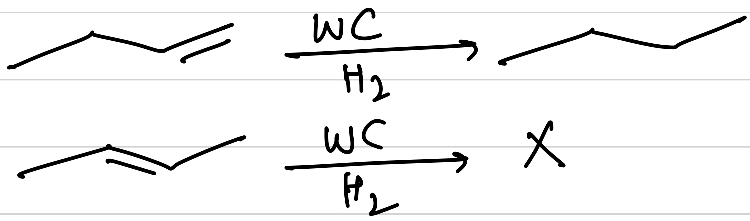
Wilkinson's catalyst $[RhCl(Ph_3P)_3]$

↓

hydrogenation of alkenes.

↓

double bond at the end of the chain is hydrogenated but double bond in centres are not affected.



Ziegler-Natta catalyst :- heterogeneous catalyst

$[(C_2H_5)_3Al + TiCl_4]$

↓

It is used for low temp. polymerization of alkene.

for ex. HDPE → high density polyethene from ethene.

cis-platin :- $[Pt(NH_3)_2Cl_2]$

↓

antitumor agent → treatment of cancer.

* Lead poisoning in the body can be removed by EDTA in the form of $[CaH_2EDTA]$

Haemoglobin → Fe^{+2} → oxyhemoglobin → Fe^{+2}

Chlorophyll → Mg^{+2}

vitamin B_{12} → Co^{+2}.